

A background image featuring a complex network diagram with numerous nodes and connecting lines, set against a dark purple gradient. The nodes are represented by small white and grey dots, and the lines are thin white lines of varying lengths, creating a web-like structure.

Data Aggregation, Privacy Threats and Anonymity

WSPC, Chapter 8

Privacy concerns

The technologies of the Internet and the Web was designed to **transfer information, not protect the privacy or security** of people who use this information;

Moreover, existing technologies provide with opportunities to collect a lot of information about users of computer networks

Example: a web site may collect information about people seeing their messages:

- Where they live; what other web sites the person has visited; their email addresses; etc



Information privacy

- Alan Westin (1967), **Information privacy**:
- “the claim of individuals, groups, or institutions to determine for themselves when, how, and to what extent information about them is communicated to others.”
- **Information privacy threat**: many individuals and institutions have lost ability to control how and to what extent information about them is communicated to marketing companies, government agencies, etc



Personal and Private Information

- **Personal information.** Information about a person, like a name, date of birth, names of parents, attended schools, etc
- **Private information.** Personal information that is not generally known. Some kinds of private information, for example bank records are protected by law.
- **GDPR (implemented 2018).** Unless a data subject has provided informed consent to data processing for one or more purposes, personal data may not be processed unless there is at least one legal basis to do so.

Personally identifiable and Anonymized information

Personally identifiable information. Information from which a person identity can be derived. **Example:** an account number.

Anonymized information. Information from which a person identity cannot be derived. Example: an age of person (if no other information is available)

Aggregate information

- Statistical information combined from many individuals to form a single record.
Example: National Census Bureau.
- No person identity can be extracted from aggregate information alone, but when combined with other anonymized information, aggregate information can help to identify and reveal particular characteristics of an individual.



Privacy threats based on anonymized information

Combining seemingly anonymous information one may reveal identity with high probability:

If you ask a person for

- a birthday
- postcode
- an age

You ask for personally identifiable information, even though it look anonymized.

Online examples:

- Screen resolution
- Battery % remaining / power available
- Operating system
- Browser & version

Open-Source Intelligence Gathering (OSINT)

- The collection and analysis of information that is gathered from public, or open, sources.
- Information can also be considered open source if it is:
 - Published or broadcast for a public audience (for example, news media content)
 - Available to the public by request (for example, census data)
 - Available to the public by subscription or purchase (for example, industry journals)
 - Could be seen or heard by any casual observer
 - Made available at a meeting open to the public
 - Obtained by visiting any place or attending any event that is open to the public

“Personal Data for the Taking”

- Title of the article in the *New Your Times*, by Tom Zeller, Jr, published May 18, 2005
- It analyses the project at John Hopkins University, done by Prof. Aviel D. Rubin and his students
- Several groups of three to four students set out to collect personal information on citizens of Baltimore using only legal, public sources of information, such as death records, property tax, campaign donations, occupational registries, etc
- Each group could spend no more that \$50
- Altogether they gathered over a million records with hundreds of thousands of individuals

“Personal Data for the Taking”(cont)

- Databases they collected were cleaned and linked, making it possible to query a multiple layers of information about a single name
- Typical result about an individual:
 - Name
 - Precise address
 - Occupation (with the details professional license)
 - The name of their spouse
 - Their birth dates
 - The price of the home they paid
 - The party registration
 - Elections he has voted in last 25 years



“Personal Data for the Taking”(cont.)

Lessons:

A lot of personal information easily available on the Web

When consolidated it may violate privacy of individuals and be used e.g. for stealing identity

Competing social interests in openness and privacy

No ready solution

A hand is holding a large, 3D '@' symbol made of cardboard on a wooden surface. The symbol is composed of concentric rings, with the outermost ring being the largest and the innermost being the smallest. The hand is positioned on the left side of the frame, with fingers visible. The background is a blurred wooden surface.

Sources of personal and private information

- User-provided information
 - When using online services or making purchases online, users provide personal information such as their names, addresses, passwords (including additional security passwords), dates of birth, and other sensitive details.
- Information obtained through the observation of user activities or from the digital traces they leave behind.
 - Log Files: Web logs, Mail Logs, DNS logs etc
 - Cookies
 - Web Bugs, Adware, Spyware



Anonymity in communications

- One of the ways to protect privacy is to make a communication **anonymous**, so an adversary that can monitor and/or compromise certain parts of the systems would not be able to match a message sender with the recipient (sender-recipient matching)
- Most widespread methods for anonymity in the Web communications based on the idea of **third trusted party** serving as **anonymizer** (special proxy server)

Anonymizer

- Early example: Anonymizer.com (many such available now)
- Upon a request from a user anonymizer fetches the web page and displays it within your browser
- In doing so, anonymizer does not leave information about your request on the web-server: it re-directs your request, replacing all sensitive information (IP address, etc) with its own details
- Additionally, it may provide
 - encryption of traffic between a user and itself
 - Blocking and removing potential active privacy and security threats: web bugs, spyware, viruses, etc



Anonymizer



Good protection and reasonable cost

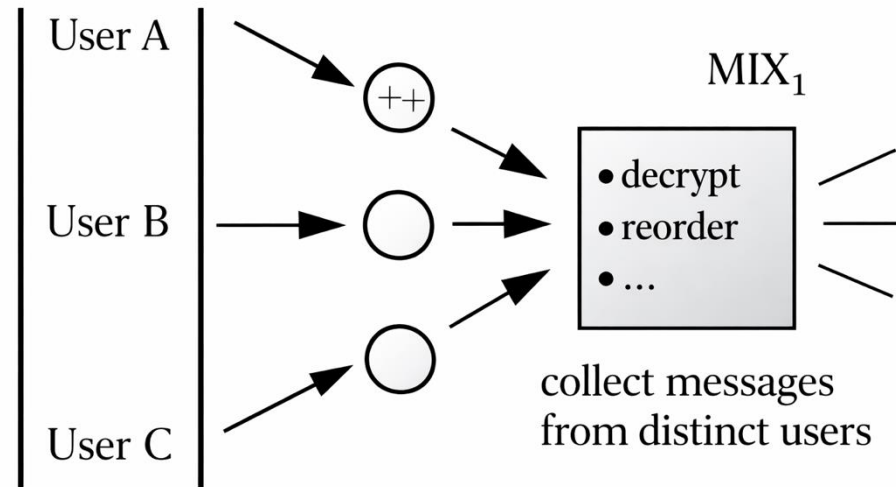


Privacy protection is based on the trusted central proxy



Central proxy “knows everything” about communication – attacks by “insiders” are possible

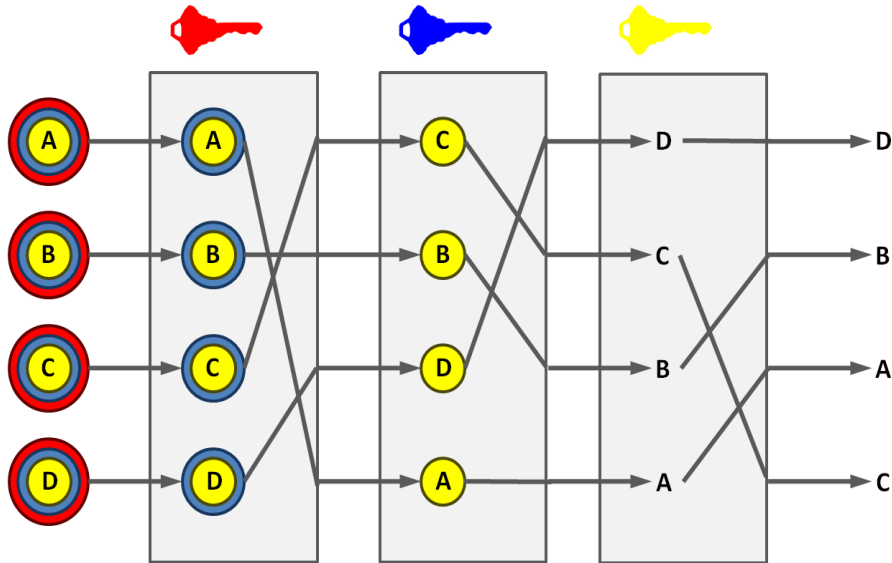
Mix- Networks



- Proposed by D. Chaum (1981)
- A **mix node** is a node in the network that takes a number of incoming messages (packets), modifies them and output in a **random order**.
- The mix nodes can be used for **anonymous** communication.

Mix-Networks

It operates as follows:



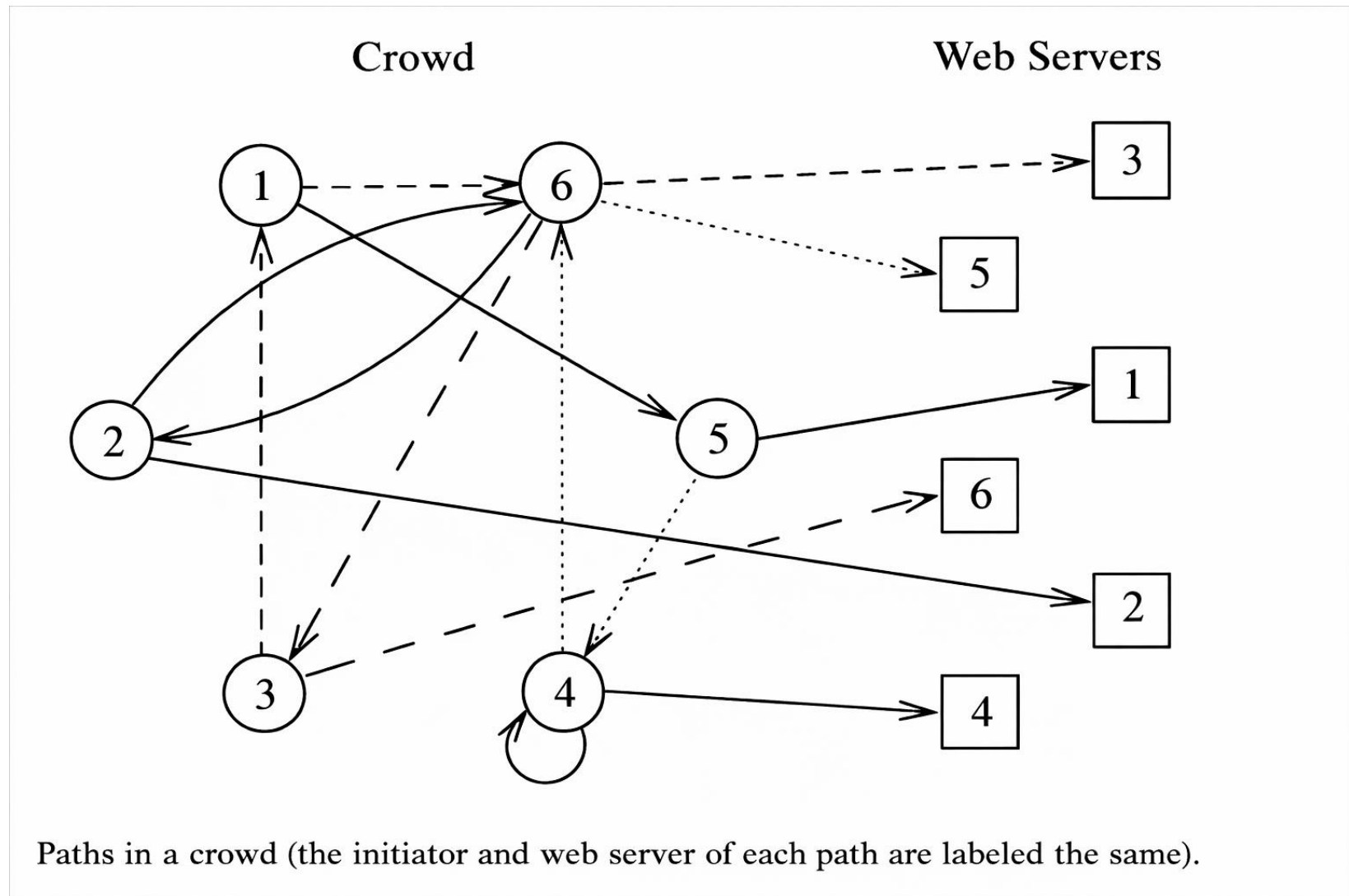
- The message is sent through a sequence of mix nodes (a route) $p_1, p_2, p_3, \dots, p_d$. The user encrypts the message with node p_d key, the result encrypt with the node p_{d-1} key, etc.
- Every mix node receives several messages, decrypt them, re-order and send to the next nodes in the route
- Every nodes “knows” only previous and the next node in the route => compromising a single, or even several (not all) mix nodes does not give an attacker an information about sender-receiver matching
- More expensive than anonymizer solution but gives more privacy protection.

Anonymity, further approaches

- M. Reiter, A. Rubin, 1998, *Crowds: Anonymity for Web Transactions*
- Based on the idea “**blending into a crowd**”, that is hiding one’s actions within the actions of many others
- To execute a web transaction a user first joins a “crowd” of other users
- Then the user’s request to a web server is passed to a random member of the crowd
- That member can either **submit the request** to the server or **forward it to another randomly chosen member** of the crowd and so on

Crowds

Paths in a crowd. Picture
by M.Reiter and A. Rubin



Privacy protection by the crowd

When the request submitted to the end server, it is submitted by a **random** member of a crowd, so identity of an initiator is hidden (“in the crowd”) from an external observer

Members of the crowd cannot identify initiator as well, they “just passing requests”



Crowds vs anonymizers and mixes



Unlike an anonymizer crowds provide no single point, where an attacker can compromise anonymity of all users



Crowds does not provide anonymity against a global adversary able to oversee all communications. In contrast, mix networks protect anonymity in that case.

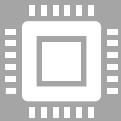


Crowds admit very efficient implementations in comparison with mixes: no encryption/ decryption operations, no inflation of message lengths.

Tor: anonymity online



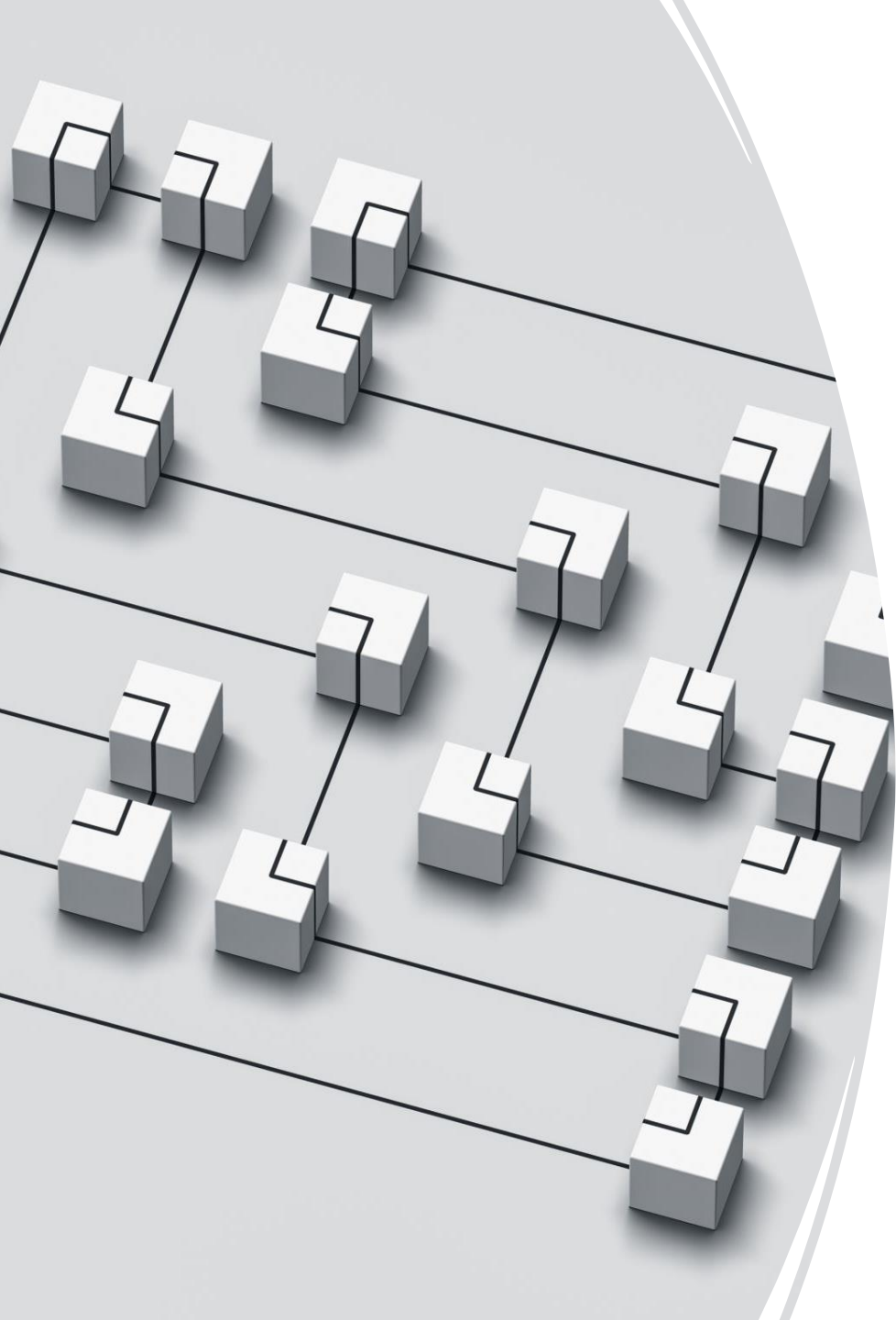
Tor (from The onion routing project, originated in the US Naval Research Lab) – practical solution for anonymity protection : www.torproject.org



It is free and open source and available for major platforms: Windows, Mac, Linux/Unix, Android



Can be used for web browsing and instant messaging, prevents people from learning your location or browsing habits



Tor: main principles

A combination of (variants of) mix-network and crowd mechanisms

- Using a set of relay nodes (crowd); currently >7000
- Routing using random choice (similar to **crowds**)
- Encrypted connections between neighbouring nodes (similar to **mix-networks**)
- It uses **public key** cryptography to share the secret keys between neighbouring nodes => uses the shared secret to perform **symmetric encryption** in further communications between neighbours
- temporarily available two-way virtual channels

Tor-Networks

- Traffic routed via encrypted nodes. Each node decrypts one layer of encryption
- Entry point is only known by the first link, and exit point is only known by the last link
- Nodes are only aware of the next link.

